FINAL REPORT — ISLES‘24 hypothesis program and the VERITAS evaluation study

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Provenance. This document is assembled **read-only** from existing verified artifacts. No VERITAS run, no SLURM job, and no statistic was computed for it; the only executions performed in its preparation were file reads, sha256 hashes, and float comparisons for the eight verification checks in §V. Ground truth is `docs/eval/GROUND_TRUTH_FROZEN.json`, sha256 `c4a349006ea887c4906b469ec10772c952ec41246e1ca9b98bd729ce87a385b9` (re-verified 2026-08-20), the authoritative six-hypothesis set {G1, H7, HYP16, G2, HYP21, G5}. Every number in this report carries a resolvable artifact path, inline or in the Appendix. Nothing here modifies `docs/eval/EVAL_SYNTHESIS.md`, `docs/eval/EVAL_RESULTS.md`, or any frozen artifact.

ABSTRACT

P(correct evidence label | correct statistic) = 3/14 = 21.4% [7.6–47.6]: of the fourteen VERITAS runs that reproduced the independently verified harness statistic — most to 15–17 significant figures — only three received the correct evidence label (`docs/eval/EVAL_RESULTS.md` §0; `docs/eval/RUNS.jsonl` + `docs/eval/FABRICATION_RESCORE.json`). **Nineteen of fifty runs (38%) carry a label and a statistic that disagree about whether the run succeeded** — eleven right-statistic/wrong-label, eight right-label/wrong-statistic, including four Config A `G1` replicates whose *correct* SUPPORTED labels rest on a double-squared residual emitting $\Delta R^2 = 103.89$ (`docs/eval/EVAL_RESULTS.md` §0). `HYP21`, the sixth frozen hypothesis, was not executable through VERITAS — its predictor is a sidecar file absent from all 79 manifest metadata keys — and contributed **0 of the 50 runs**; the realised design is therefore **5 hypotheses × 5 replicates × 2 model configurations = 50 runs**, not the frozen 6×5×2 = 60 (`docs/eval/EVAL_SYNTHESIS.md` §8). Upgrading the Phase-2B coding role ~10× in active parameters raised statistic reproduction from 8.0% to 48.0% of launched runs and *lowered* evidence-label accuracy from 42.1% to 15.0% — below that arm’s own trivial always-REFUTED baseline of 20.0% — with every one of the eleven right-statistic/wrong-label failures located in deterministic framework code that executes after the language model has finished. On the scientific side, the project’s verified positives are exactly two (G1, H1); the mediation thesis survived a pre-registered falsification batch built to break it; and the one apparent new positive (HYP16) is voided by its own pre-registered secondary analysis.

PART I — SCIENTIFIC RESULTS

I.1 Verified positives: exactly two

The only verified positives in the project are `G1` and `H1`. This is recorded in the frozen ground truth's `tier2_exclusion.verified_positives_in_project` block (`docs/eval/GROUND_TRUTH_FROZEN.json`), in `docs/eval/APPENDIX.md:59–67`, and in `docs/eval/EVAL_PROGRESS.md:185–187`.

- **`G1` — NIHSS at 24 h over Age + admission NIHSS → 90-day mRS.** ΔR^2 = **0.2567** (0.25669474105632395), partial F = 31.67, p = 3.76e-07, n = 72; Holm m=1, survives; ≥ SESOI. Artifact: `results/codex_v3/jobs/1012517/G1/statistics.json` (sha256 in the frozen file); passed all seven reproduction gates (`docs/eval/APPENDIX.md` §A1). One stated caveat: π@SESOI = 0.7963, marginally below the study's own 0.80 threshold — G1 is the strongest verified positive and must not be called well-powered (`docs/eval/EVAL_SYNTHESIS.md` §8).
- **`H1` — premorbid mRS over Age + admission NIHSS → 90-day mRS.** ΔR^2 = **0.1513** (0.15127797845775737), p = 9.49e-05, n = 78; Holm m=3 → 2.85e-04, survives; ≥ SESOI (`docs/HYPOTHESIS_CENSUS.md:61`; artifact `results/codex_v2/jobs/1008042/H1/statistics.json`). Two qualifications, both pre-registered: complete-case n = 78 of the 129-subject cohort — **≈40% of subjects lost to missingness**, dominated by the premorbid-mRS field itself, populated for 80 of 129 (`docs/PREREG_V2.md:73`, "Effective n: 78 (complete-case, excl 9999)"; `docs/codex_hypotheses_v2.md:273`); and H1 is the batch's expected positive — premorbid disability is a textbook determinant of 90-day mRS (`docs/codex_hypotheses_v2.md:139`) — so it functions as the cohort's **positive control**, not a discovery.

`isles_07` (ΔR^2 = 0.149) is Tier 2 and excluded from the verified positives, despite being listed as the closed keystone in `AGENTS.md`. Three independent disqualifiers, recorded at `docs/eval/APPENDIX.md:59–67` and `docs/eval/EVAL_PROGRESS.md:185–187` and frozen in `GROUND_TRUTH_FROZEN.json.tier2_exclusion`:

1. **No `partial_f`** in any of the 54 bank `ground_truth.statistics` blocks — only `effect_size` and `p_value` — so the harness-verified ΔR^2/F/p criterion cannot be met from the bank at all.
2. **Provenance**: 19 of 29 batch-A rows carry `dataset_derived_RECONSTRUCTED` — rebuilt after the 2026-07-01 bank loss, a reconstruction known to have introduced at least one wrong value (the `isles_42` expected label, corrected 2026-07-10).
3. **Artifact granularity**: directory-level paths (17/29 point at `results/veritas_main/` with a document-section reference); no per-hypothesis `statistics.json` with design matrix, residuals and ordered subject ids exists, so independent recomputation is impossible.

Tier 2 does not mean wrong — 14 batch-A rows have runnable generator scripts and `isles_13`/`isles_14` are flagged AUTHORITATIVE — it means **not independently verifiable from disk today**. Promotion to Tier 1 requires re-running the generators, which is compute, and compute is frozen (`GROUND_TRUTH_FROZEN.json.tier2_exclusion.promotable_subset_note`).

I.2 The mediation thesis

The program's organising scientific claim (`docs/BATCH_HYP11–20_SYNTHESIS.md` §1):

> *Lesion-intrinsic features are null because infarct volume already carries their signal; frailty features are null because age already carries theirs.*

Its sharpest surviving formulation, after the falsification batch (`docs/BATCH_HYP11–20_SYNTHESIS.md` §5): once total infarct volume is known, *where* the infarct is adds at most a small increment, and what increment exists is concentrated in the corticospinal tract rather than distributed across a broad eloquent-tissue union — ~1/3 of SESOI, not surviving family-wise correction in this cohort. Lane (b), lesion location, has had four independent attempts: `isles_50` (eloquent-union fraction, $\Delta R^2 = 0.0020$), `HYP11` (union volume, 0.00058), `HYP12` (CST, 0.0334), `HYP21` (PLIC, 0.0023) — "lesion location adds nothing decisive beyond lesion volume" (`docs/BATCH_HYP11–20_RESULTS.md`, "What survives" section). The volume→outcome signal the thesis rests on is historically anchored by `isles_07` ($\Delta R^2 = 0.1486$, $p = 9.0\text{e-}7$, `docs/HYPOTHESIS_CENSUS.md:30`) — which, per §I.1, is Tier 2: consistent with everything measured since, but not independently recomputable from disk.

I.3 The pre-registered falsification

Batch HYP11–20 was designed to put the thesis at genuine risk: ten hypotheses, **seven of them pre-declared mediation-resistant (LOW mediation risk)** — reserve/substrate (HYP13, 14, 15), process (HYP17), discordance (HYP18), and effect modification (HYP19, HYP20) — run against SESOI $\Delta R^2 = 0.10$ with pre-registration locked and sha-verified before the first statistic (`docs/BATCH_HYP11–20_SYNTHESIS.md` §1; prereg `docs/prereg/BATCH_HYP11–20.md`).

All seven returned well-powered nulls under Holm, at power 0.85–0.96 (`docs/BATCH_HYP11–20_SYNTHESIS.md` §1, §5; per-hypothesis artifacts `work/hyp11_20/jobs/*/HYP*/ statistics.json`, independently re-verified from disk — 16/16 packages, 27/27 checks, job 1014397, `docs/HYPOTHESIS_CENSUS.md` §4). None produced an effect within a factor of three of SESOI. The thesis survived a test built to break it; the informative nulls (HYP19's age×volume interaction, HYP20's collateral×volume interaction, HYP18's discordance, HYP17's time-to-recanalization) are itemised in `docs/BATCH_HYP11–20_SYNTHESIS.md` §2.

I.4 HYP21: the ordered prediction failed on its own terms

`HYP21` (PLIC-restricted infarct fraction over Age + NIHSS + volume → 90-day mRS) carried a pre-registered **ordered prediction**: if HYP12's corticospinal-tract signal is driven by the motor-tract bottleneck, restricting the mask to the posterior limb of the internal capsule should *concentrate* it. It did the opposite: $\Delta R^2 =$ **0.0023** (0.002312939409511383), $p = 0.5205$, power 0.960 at SESOI (`work/hyp21/jobs/1013974/HYP21/statistics.json`; `docs/HYPOTHESIS_CENSUS.md:83`) against HYP12's

0.0334. The refutation is of the **ordering of point estimates**, deliberately pre-specified that way because the cohort cannot support a formal test of the difference between two ΔR^2 on the same subjects — so it fails on its own terms, independent of significance (`docs/BATCH_HYP11–20_RESULTS.md`, "What survives" section).

Caveat, measured before the test: only 25.5% of the PLIC lies inside HYP12's CST mask (`docs/BATCH_HYP11–20_RESULTS.md`; `docs/HYPOTHESIS_CENSUS.md:152`). The two masks are largely different tissue — PLIC is not an anatomical subset of the CST mask as the lane's framing might suggest — so the CST signal is evidently in the non-capsular portion (corona radiata / centrum semiovale). That reading is post-hoc speculation about a refuted hypothesis, not a finding. Two further constraints: power at $\Delta R^2 = 0.05$ with $n = 126$ is only 0.724 (the null is well-powered *at SESOI 0.10* only, and is not strong evidence against a small PLIC effect), and 59/129 subjects have zero PLIC involvement (median fraction 0.00205), limiting what the test could detect.

I.5 HYP12: the live lead, and what it needs

`HYP12` (corticospinal-tract lesion load) is the one place lane (b) is not a clean sweep: $\Delta R^2 = 0.0334$, raw $p = 0.0135$, **correct direction**, Holm $p = 0.122$ (`work/hyp11_20/jobs/1013910/HYP12/statistics.json`; `docs/HYPOTHESIS_CENSUS.md:74`). The ordering `isles_50` (0.0020) → `HYP11` (0.00058) → `HYP12` (0.0334) is exactly what diluting a real tract-specific effect across a large union mask would produce (`docs/BATCH_HYP11–20_SYNTHESIS.md` §4). It is a **lead, not a finding**: 3× below SESOI and Holm-dead in this cohort. The pre-registered power arithmetic says what it needs: power at $\Delta R^2 = 0.05$ with $n = 126$ is 0.724; **n = 152 would be needed for 0.80** (shortfall 26), and the effect actually at stake (~0.033) is below even that (`docs/BATCH_HYP11–20_RESULTS.md:157–159`). ISLES'24 is exhausted at $n \approx 126$; HYP12 needs an external cohort.

I.6 HYP16: the methods demonstration

`HYP16` (penumbral salvage fraction → 90-day mRS) cleared Holm (adjusted $p = 0.00045$) and cleared SESOI ($\Delta R^2 = 0.10811213368156236$, $p = 4.47\text{e-}05$; `work/hyp11_20/jobs/1013907/HYP16/statistics.json`) — and is **scientifically void, by its own pre-registration**. The exposure embeds final infarct volume by construction, and the prereg said before the number existed that a positive here is weak evidence and the volume-adjusted secondary is the decisive model. That secondary removes the entire effect: **ΔR^2 0.108 → 0.0015 (0.001536), p 4.5e-5 → 0.605** (`work/hyp11_20/jobs/1013907/HYP16_SECONDARY/`; `docs/HYPOTHESIS_CENSUS.md:93`; `docs/BATCH_HYP11–20_SYNTHESIS.md` §3). The POSITIVE label is retained beside its refutation because the decision rule was fixed in advance and retro-editing a rule to erase an inconvenient label is the failure mode this project exists to prevent. As a methods demonstration it is the project's cleanest: **the correct scientific answer lived in a pre-registered secondary that no label machinery consulted** — which is precisely why HYP16 was chosen as the hidden-covariate divergence cell of the Part II ground-truth set.

I.7 Limits on Part I

- **~97% reperfusion removes the target variance.** ISLES'24 is a ~97%-reperfused cohort (`docs/codex_hypotheses_v2.md:68,275`; `docs/codex_hypotheses_v3.md:30`): TICI is variance-limited, and features whose mechanism runs through reperfusion success have almost no outcome variance left to explain. Nulls in that lane are conditional on this cohort's treatment profile and must not be generalised to untreated or poorly-reperfused populations.
- **SESOI 0.10 does not exclude the 0.02–0.05 band.** Every null here is a null *at SESOI 0.10*. The record contains several reproducible sub-SESOI effects in the $\Delta R^2 \approx 0.02$–0.05 band — HYP12 0.0334, HYP19 0.0329, HYP14 0.0314 (direction-reversing), isles_29 0.0439, isles_46 0.0365, G5 0.0460 (wrong-signed) (`docs/BATCH_HYP11–20_SYNTHESIS.md` §5; `docs/HYPOTHESIS_CENSUS.md` §1-2). The batch licenses "none of these is clinically decisive at the pre-set bar", not "these features are inert". Interaction tests (HYP19/20) are additionally structurally disadvantaged at a main-effect SESOI.
- **Single cohort, no external validation.** Every result above is single-site, $n \approx 129$, one imaging pipeline, one extraction stack. Nothing in Part I has been tested on data this project did not process. Effective multiplicity is also understated by design: the ten batch-E tests share overlapping cohorts and segmentations (`docs/BATCH_HYP11–20_SYNTHESIS.md` §4).

PART II — VERITAS EVALUATION

Condensed from `docs/eval/EVAL_SYNTHESIS.md`; tables in `docs/eval/EVAL_RESULTS.md`; per-run receipts in `docs/eval/APPENDIX.md` §A10 and `docs/eval/RUNS.jsonl`.

II.1 Framing and non-comparability

This is a **domain-transfer and measurement-depth evaluation**, not a replication of arXiv:2604.12144 and not a claim its results fail to reproduce. VERITAS's published metrics measure label agreement; they do not measure whether the underlying statistics are correct. This study measured both, on an out-of-paper domain (ISLES'24, incremental-prediction ΔR^2/partial-F family), against an independently verified NumPy/SciPy harness. **No ISLES percentage here is comparable to the paper's ACDC figures** (v2 Table 4: evidence-label 91.0 / verdict 87.9): different test family, different tooling (numpy+scipy only), different cohort, half the replicates, and ISLES appears nowhere in the paper (`docs/eval/EVAL_SYNTHESIS.md` §1). What is comparable is the label machinery itself: $\alpha = 0.05$, $\pi_0 = 0.80$, Table 1's rules as implemented at `experiments/evaluation.py:1071–1085`.

II.2 Headline

14 of 50 runs reproduced the harness statistic (ΔR^2 within rtol 1e-3 and p within rtol 1e-2, measured from the Phase-2B emitted file alone, per defect D11 — tolerances are study-defined; the paper specifies none). Agreement in those fourteen runs to 15–17 significant figures; one run (`B/HYP16/r5`) bit-exact. **Eleven of the fourteen received the wrong evidence label**, giving the abstract's conditional, 3/14 = 21.4% [7.6–47.6]. All three correct labels are `G2`, all reached through the power clause; **no correct label anywhere in the study was earned through the significance path on a statistic that reproduced** (8 of 11 correct labels came via the significance path, and every one of those eight sat on wrong arithmetic — four `A/G1` double-squares, three `A/H7` including a fabricated placeholder file, one `B/G1/r5` double-square). The converse — label right, statistic wrong — fires on 7 Config A runs and 1 Config B run; on `G1` under Config A, the four replicates scored *correct* are exactly the four that double-squared (ΔR^2 = 103.89, r2_base = −193.36). **Combined: 19 of 50 runs (38%)** (`docs/eval/EVAL_RESULTS.md` §0; `docs/eval/EVAL_SYNTHESIS.md` §2).

One run deserves its own sentence. `B/G5/r3` reproduced ΔR^2 and p to 16 significant figures; its Phase-3 agent correctly wrote that the finding contradicts the hypothesis and reported REFUTED — the frozen ground truth. The evidence operator computed SUPPORTED, the conflict resolver emitted **INVALID**: the agent was right and was overruled into an unscoreable label (`results/eval/B/G5_r3/exp_20260816_053552/evaluation_results.json`).

II.3 The arm contrast

The manipulation: `qwen3-coder:30b` (MoE, ≈3.3 B active parameters) → `qwen2.5-coder:32b` (dense, 32.8 B active) in the Phase-2B `statistician_coding` role — **~10× active parameters**, never "30b vs 32b" (`docs/eval/APPENDIX.md` §A3). All prompts, seeds-policy, temperature, hardware (`gpu:h200:1`) held identical.

quantity	Config A	Config B	source
statistic reproduction / launched	8.0% (2/25)	**48.0% (12/25)**	`EVAL_RESULTS.md` T3
evidence-label accuracy	42.1% (8/19)	**15.0% (3/20)**	T2
always-REFUTED baseline, same rows	10.5%	20.0%	T2
uplift over baseline	+31.6 pp	**−5.0 pp**	T2
completion rate	60.0% (15/25)	32.0% (8/25)	T2

Config B scored below its own trivial baseline — and this is a mechanism result, not a capacity result. Ten of Config B's twenty scoreable rows carry INVALID forced by a Phase-2B input gate that fired *after* the run had computed the right numbers: §18 (`variables_tested` ≠ planned predictors) fired on 11/25, §16

(`synthetic_data_detected`) on 10/25 — of those ten, **eight operative files demonstrably read real patient data and five reproduce the harness statistic to 15–16 significant figures** (`docs/eval/EVAL_RESULTS.md` T1; `docs/eval/SYNTHETIC_FP_RECEIPTS.md`).

II.4 Four architectural defects, one property

defect	what it does	locus
absent SESOI magnitude gate	significant-at-any-magnitude → SUPPORTED; `sesoi_value` feeds only `_compute_power_at_sesoi`	`experiments/evaluation.py:1071–1085`
§15 degenerate direction clause	`direction_intent:"negative"` unsatisfiable for any ΔR^2 test	deterministic evaluator
§16 synthetic-data guard	fires on `"data"` as a substring inside `"metadata"` ±3 lines of a “placeholder” comment → INVALID	Phase-2B input gate
§18 jointly unsatisfiable contracts	Phase-1 plan demands literal field names; domain note instructs a descriptive string; validator demands they match	Phase-1/2B contract pair

All four live in deterministic Python downstream of the LLM; no model at any scale has a causal path to any of them. The ~10× Phase-2B upgrade changed none, and two got worse: §16’s false-positive rate *rises* with model quality (a better model writes richer comments — more substring surface; 0/25 firings in A, 10/25 in B), and §18 rejects a model *for obeying* the reporting instruction (`docs/eval/EVAL_SYNTHESIS.md` §4; `EVAL_RESULTS.md` T5). The asymmetry matters: the SESOI gap admits wrong answers (true-but-negligible → SUPPORTED); §15/§16/§18 reject right ones (all eleven right-statistic/wrong-label runs). §16 is over-broad, not simply broken — one of its four terminal firings is a true positive (`B/G1/r3` , `docs/eval/SYNTHETIC_FP_RECEIPTS.md`). This split — capacity-limited arithmetic defects that moved with model scale vs architectural defects that did not — is the study’s central result.

II.5 Predicted vs observed

Predictions were registered 2026-08-14 (`docs/eval/PREDICTIONS.md`), before Config B existed; staged adjudications in `docs/eval/T5_ADJUDICATION.md` ; full table `EVAL_RESULTS.md` T5.

- **Mode 1, double-square: CONFIRMED** — reduced above half, not eliminated (8/25 → 4/25 launched; 42.1% → 19.0% on the emitting base), under a domain note that already instructs manual RSS computation — the third documented occurrence under that instruction is `A/G1/r1` (`results/eval/A/G1_r1/exp_20260815_201821/…/workspace/code/analysis.py:64`).
- **Mode 5, the sharp case: numeric half confirmed to the digit; the falsifier never fired.** `B/H7/r5` (job 1027040) emitted $\Delta R^2 = 0.06896015323049387$ and $p = 0.0012904403132413423$ — the first correct `H7` statistic in the project’s history, 1.450× below SESOI — and the evidence operator computed `evidence_label_raw = SUPPORTED` exactly as pre-registered; §18 then overwrote the final label to

INVALID before it could be reported, so the sharp case remains open. *(Provenance note for this number:* `B/H7/r5` *'s emitted file carries no* `delta_r2` *key. The* ΔR^2 *quoted here and in* `EVAL_RESULTS.md` *§0 is derived as* `r2_full − r2_base` *= 0.23821738036350737 − 0.1692572271330135 = 0.06896015323049387, which equals the file's* `effect_size` *field, typed* `delta_r_squared` *— see §V check 6.)*

- **The pre-committed coupling criterion: NOT MET.** Reproduction rose 6-fold; label accuracy moved four cells *downward* — anti-correlated, a stronger decoupling than the criterion had a branch for. That is an observation about the criterion's coverage, not a rescue: the prediction is recorded not met, and the mechanism (§16/§18 input gates) is not the label logic the divergence thesis is about.
- **Mode 4 (`P2B_RESULTS_SCHEMA_INVALID`): REFUTED** — predicted reduced, went 4/25 → 14/25, for architectural (§16/§18) reasons. Mode 2 (fabrication) 1→0 on a single event; mode 3 (P1 deaths) confirmed on its own terms but for the wrong reason (§18, not plan capability). A residual arithmetic defect was characterised as pre-committed: two Config B runs emit VIF < 1 while reproducing $\Delta R^2/p$ to 15+ figures (`B/HYP16/r2` 0.008116, `B/G5/r2` 0.586).

II.6 G5 and the direction clause

`G5` is the frozen set's only REFUTED-by-opposite-direction cell, included to test exactly that clause — **and the clause failed it**. All five Config B replicates reaching the evidence operator returned SUPPORTED (three of them on exactly-reproduced statistics); the opposite-direction branch never fired. The one escape (`B/G5/r3`) came from the agent being right and getting overruled to INVALID, not from the direction rule (`docs/eval/EVAL_SYNTHESIS.md` §6; `EVAL_RESULTS.md` T4). Together with §15 (structurally unsatisfiable negative direction), the direction logic fails open in one direction and closed in the other — both in deterministic evaluator code.

II.7 The instrument, with its own defect log

`scripts/eval/score_run.py` (plus the non-mutating re-scorer `detect_fabrication.py`) was validated before use — 10/10 archive mode matches, all five documented failure modes independently re-derived (`docs/eval/INSTRUMENT_VALIDATION.json`) — and then frozen for the whole of collection. Its design principle: every classification rests on something the judged artifact cannot produce (double-square by independent numeric identity, fabrication by precision test, reproduction from the emitted file only). **Fifteen defects (D1–D15) are logged across 2,784 lines of study code**, three found only because live data contradicted the instrument, two of those directly beneath the headline metric (D11: scorecard narrative preferred over computed output; D14: the instrument committed the exact operative-file error it was built to catch). Three remain open: D6 (by decision), D7 (awaiting adjudication, §III), D15 (deliberately unfixed, disclosed). Full log: `docs/eval/EVAL_PROGRESS.md` § INSTRUMENT DEFECT LOG; `EVAL_RESULTS.md` T6.

II.8 Limitations of Part II

Verbatim summary of `docs/eval/EVAL_SYNTHESIS.md` §8: `HYP21` not executable (0/50; sidecar predictor unreachable through the only shared data path). Pilot scale — every rate on $n \le 25$, the widest CI spans 56 points; **no rate is a population estimate**. Non-comparability to the paper, itemised. Replicates unseeded because the framework exposes no seed parameter (`src/veritas/langchain_utils.py:163–191`) — required for the T4 nondeterminism measurement (9 of 10 hypothesis×arm cells produced >1 distinct label across 5 replicates), but replicate and sampling variance cannot be separated. `G1` marginally under-powered (0.7963). `isles_29`'s numbers transcribed, not gate-verified. Statistic tolerances study-defined. `HYP16`'s prompt authored during the study (D12), its Phase-1 rate reported separately. **The headline conditional 3/14 is an observation, not a pre-registered test**, and must not be quoted with the authority of one — its value is that all 14 rows carry artifact paths and code-located mechanisms.

PART III — OPEN QUESTIONS

III.1 Scientific

1. **Does the corticospinal-tract signal replicate?** `HYP12`, the one live lead (§I.5) — $\Delta R^2 = 0.0334$, correct direction, Holm $p = 0.122$ — cannot be advanced on ISLES'24: the cohort is exhausted at $n \approx 126$, power at $\Delta R^2 = 0.05$ is 0.724, and $n = 152$ would be required for 0.80 (`docs/BATCH_HYP11–20_RESULTS.md:157–159`), with the effect actually at stake (~0.033) below even that. Whether the CST-lesion-load signal replicates in an external cohort of $n \ge 152$ is open, and cannot be answered from this dataset.
2. **Do the Tier-2 values survive independent regeneration?** 14 batch-A rows have runnable generator scripts; `isles_13`/`isles_14` are flagged AUTHORITATIVE. Re-running the generators would either promote those rows — including the keystone `isles_07` ($\Delta R^2 = 0.149$) — to Tier-1 verified status, or surface further reconstruction errors of the `isles_42` kind. The regeneration has not been run — it is compute, and compute is frozen (`GROUND_TRUTH_FROZEN.json.tier2_exclusion.promotable_subset_note`) — so the keystone's independent verifiability remains open.

III.2 Evaluation framework

1. **The SESOI gap** (framework defect notes §2; `experiments/evaluation.py:1071–1085`). Table 1 has no magnitude clause: `sesoi_value` is computed and used only for power, never in the `p < α` branch, so `H7` ($\Delta R^2 = 0.0690$, 1.45× below SESOI) labels SUPPORTED, and the scorer's "standard" profile hardcodes `sesoi_value = 0.3` against a pre-registered 0.10 (framework defect notes §11b). Open: whether the label vocabulary should gain a DETECTED_BUT_SUB_SESOI (or equivalent) state; where SESOI should

come from per-hypothesis; and what the vocabulary should do with `isles_29`-class cases, where the bootstrap CI spans the SESOI and **no correct label exists** (`docs/eval/APPENDIX.md` §A1, control row).

2. **The `adjusted_for` self-report gate** (framework defect notes §4; `evaluation.py:1004–1013`). For required-`adjust_for` hypotheses the evaluator checks only the emitted `phase2b.adjusted_for` key and never consults `variables_tested.predictors`, where the model actually records its covariates — so correctly adjusted runs (isles_40, all three replicates) score INVALID on a contract technicality. Two remedies are available, separately or together — auto-emitting `adjusted_for` from the fitted covariate set, or falling back to `variables_tested.predictors` before declaring a violation — and which the contract should adopt is open.
3. **§16 breadth** (framework defect notes §16; `phase2b_statistics.py:1111–1121`). The synthetic-data guard matches prose — `"data"` as a substring inside `"metadata"` near a "placeholder" comment — and its harm grows with model quality (0/25 firings under Config A, 10/25 under Config B; 8 of the 10 read real data, 5 reproduce ground truth to 15–16 figures; receipts `docs/eval/SYNTHETIC_FP_RECEIPTS.md`). It also catches one genuine fabrication (`B/G1/r3`), so the open question is not whether the guard should exist but whether detection should be behavioural (does the operative file access data; do the emitted values reproduce an independent quantity) rather than lexical — the same criterion this study's D14 fix converged on.
4. **D7 — `literal_p` adjudication** (framework defect notes §14; defect log `docs/eval/EVAL_PROGRESS.md` D7). On `G6` the study instrument reports `literal_p = True` on the operative file (`analysis_v6.py:105`, `p_value = 0.05` assigned as the result) while the framework's `phase2b_literal_pvalue_assignment` reports False. Line 105 matches App. E.6's own definition, so the framework appears to under-detect — but the disagreement is unadjudicated; which detector is right is open.
5. **The mode-5 sharp case.** The SESOI-gate question at the *reported*-label level — whether a sub-SESOI SUPPORTED label survives to the final report — was left open only because §18 fired first (§II.5). Answering it requires a hypothesis whose tested term is a raw manifest field, so that §18 cannot fire (`docs/eval/EVAL_SYNTHESIS.md` §5); no such run family exists in the present record.
6. **How much of the label nondeterminism is sampling variance?** The framework exposes no LLM seed parameter (`src/veritas/langchain_utils.py:163–191`), so replicate variance and sampling variance are inseparable as shipped (§II.8): 9 of 10 hypothesis×arm cells produced more than one distinct label across five replicates, but partitioning that variability is impossible without seed control the framework does not currently expose.

Scope note. Per the batch synthesis (`docs/BATCH_HYP11–20_SYNTHESIS.md` §6), nothing in HYP11–21 qualifies as a positive finding; the two verified positives (§I.1) remain the only results recorded as such, each with its stated caveat.

V — VERIFICATION AND DISCLOSURES

Disclosure 1 — eight numeric claims were traced to disk, not all. In preparing this report, exactly **eight** load-bearing numeric claims were re-verified against their on-disk artifacts (table below, re-run 2026-08-20 against this document). Every other number in this report is transcribed from the named, previously verified documents (`docs/eval/EVAL_RESULTS.md`, `docs/eval/EVAL_SYNTHESIS.md`, `docs/eval/APPENDIX.md`, `docs/eval/EVAL_PROGRESS.md`, `docs/HYPOTHESIS_CENSUS.md`, `docs/BATCH_HYP11–20_{SYNTHESIS,RESULTS}.md`, the framework defect notes) with the path cited in place, and was **not** independently re-opened at the artifact level for this report.

Disclosure 2 — B/H7/r5's ΔR² is derived, not emitted under that name. The emitted file `results/eval/B/H7_r5/exp_20260816_014412/isles_59_large_core_threshold_mrs3m_increment/run_000/phase2b_statistical_analysis/workspace/data/statistical_results.json` contains **no `delta_r2` key**. The reported ΔR² = 0.06896015323049387 is derived as `r2_full − r2_base` (0.23821738036350737 − 0.1692572271330135); the file's `effect_size` field (typed `delta_r_squared`) carries the same value. This applies everywhere that number is used (§II.5 mode 5; `EVAL_RESULTS.md` §0 row 4 and T5).

The eight checks. Gate (precondition): sha256 of `docs/eval/GROUND_TRUTH_FROZEN.json` equals `c4a349006ea887c4906b469ec10772c952ec41246e1ca9b98bd729ce87a385b9` — **PASS**.

#	claim in this report	artifact	result
1	G1: ΔR² = 0.25669474105632395, p = 3.75995859999778e-07, n = 72	`results/codex_v3/jobs/1012517/G1/statistics.json`	**PASS**
2	H1: ΔR² = 0.15127797845775737, p = 9.493756773005297e-05, n = 78	`results/codex_v2/jobs/1008042/H1/statistics.json`	**PASS**
3	H7 (GT): ΔR² = 0.06896015323049376, p = 0.0012904403132413533, power@SESOI = 0.9574663450150511	`results/codex_v2/jobs/1008054/H7/statistics.json`	**PASS**
4	HYP16 primary: ΔR² = 0.10811213368156236, p = 4.4669592351209874e-05	`work/hyp11_20/jobs/1013907/HYP16/statistics.json`	**PASS**
5	HYP21: ΔR² = 0.002312939409511383, p = 0.5205449888878269, power = 0.9600988657422707; and 0 of 50 runs	`work/hyp21/jobs/1013974/HYP21/statistics.json`; `docs/eval/RUNS.jsonl`	**PASS**
6	B/H7/r5 emitted: no `delta_r2` key; `r2_full − r2_base` = 0.06896015323049387 = `effect_size`; p = 0.0012904403132413423	`results/eval/B/H7_r5/…/statistical_results.json` (full path above)	**PASS**
7	50 runs; 14 reproduce (D11-corrected); 11 of 14 wrong label; the 3 correct all `G2` via power clause; 21.4% [7.6–47.6] Wilson	`docs/eval/RUNS.jsonl` + `docs/eval/FABRICATION_RESCORE.json`	**PASS**

#	claim in this report	artifact	result
8	Converse divergence 8 (7 A + 1 B); combined 19 of 50	`docs/eval/RUNS.jsonl` + `docs/eval/FABRICATION_RESCORE.json`	**PASS**

Method: direct JSON reads and float equality (exact on stored values; Wilson interval recomputed at z = 1.96 → [0.0757, 0.4759], rounding to the quoted [7.6–47.6]). Checks executed read-only; no artifact was modified.

APPENDIX — artifact paths

All paths relative to the project root.

Frozen ground truth and per-hypothesis harness artifacts (sha256 for each artifact is recorded inside the frozen file):

item	path
Frozen ground truth (sha `c4a34900…385b9`, mode 444)	`docs/eval/GROUND_TRUTH_FROZEN.json`
G1	`results/codex_v3/jobs/1012517/G1/statistics.json`
H7	`results/codex_v2/jobs/1008054/H7/statistics.json`
HYP16 (primary)	`work/hyp11_20/jobs/1013907/HYP16/statistics.json`
HYP16 (pre-registered secondary)	`work/hyp11_20/jobs/1013907/HYP16_SECONDARY/`
G2	`results/codex_v3/jobs/1012523/G2/statistics.json`
HYP21	`work/hyp21/jobs/1013974/HYP21/statistics.json`
G5	`results/codex_v3/jobs/1012537/G5/statistics.json`
H1 (verified positive, not in frozen set)	`results/codex_v2/jobs/1008042/H1/statistics.json`
HYP12 (live lead)	`work/hyp11_20/jobs/1013910/HYP12/statistics.json`
isles_29 control (narrative only, no statistics.json)	`isles_29_exploratory_characterization/`

item	path
Pinned manifest (sha `c4fad070…acdbe`)	`data/isles_manifest/dataset_manifest.json`

Evaluation study record:

item	path
Spec (authoritative)	`docs/eval/CLAUDE_EVAL.md`
State ledger + defect log D1–D15	`docs/eval/EVAL_PROGRESS.md`
Tables T1–T6	`docs/eval/EVAL_RESULTS.md`
Synthesis	`docs/eval/EVAL_SYNTHESIS.md`
Supporting appendix A1–A10 (incl. all 50 per-run receipts)	`docs/eval/APPENDIX.md`
Machine-readable run record (50 runs)	`docs/eval/RUNS.jsonl`
D10/D11/D14 re-score of all 50 runs	`docs/eval/FABRICATION_RESCORE.json`
Instrument validation gate (10/10)	`docs/eval/INSTRUMENT_VALIDATION.json`
Pre-registered predictions (signed 2026-08-14)	`docs/eval/PREDICTIONS.md`
Staged T5 adjudications	`docs/eval/T5_ADJUDICATION.md`
§16 firing receipts and adjudication	`docs/eval/SYNTHETIC_FP_RECEIPTS.md`, `docs/eval/SYNTHETIC_FP_AUDIT.json`, `docs/eval/SYNTHETIC_FP_RECEIPTS.json`
Quarantined D12 runs (in no rate)	`docs/eval/RUNS_QUARANTINED_invalid_prompt.jsonl`
Study scripts (13 scripts, 2,784 lines)	`scripts/eval/*.py`
Reference PDF (sha `792814ed…d975`)	`docs/reference/veritas_arxiv_2604.12144v2.pdf`
Per-run emitted statistics (any run)	`<run_dir>/<bank_id>/run_000/phase2b_statistical_analysis/workspace/data/statistical_results.json`
B/H7/r5 emitted file (Disclosure 2)	`results/eval/B/H7_r5/exp_20260816_014412/isles_59_large_core_threshold_mrs3m_increment/run_000/phase2b_statistical_analysis/workspace/data/statistical_results.json`

item	path
A/G1/r1 double-square receipt	`results/eval/A/G1_r1/exp_20260815_201821/…/workspace/code/analysis.py:64`
B/G5/r3 agent-overruled receipt	`results/eval/B/G5_r3/exp_20260816_053552/evaluation_results.json`

Scientific program record:

item	path
Hypothesis census (all batches, aliases, caveats)	`docs/HYPOTHESIS_CENSUS.md`
Batch HYP11–20 synthesis (mediation thesis test)	`docs/BATCH_HYP11-20_SYNTHESIS.md`
Batch HYP11–20 / HYP21 full numbers	`docs/BATCH_HYP11-20_RESULTS.md`
Locked pre-registration, batch C (H1–H7)	`docs/PREREG_V2.md`
Hypothesis authoring docs (cohort facts, ~97% reperfusion)	`docs/codex_hypotheses_v2.md`, `docs/codex_hypotheses_v3.md`
ISLES sweep final status (incl. isles_29 source lines 70–87)	`docs/ISLES_sweep_final_status.md`
Batch E/F disk re-verification (16/16 packages, job 1014397)	`docs/HYPOTHESIS_CENSUS.md` §4

HALTs: none, at any point in the program (`docs/eval/EVAL_PROGRESS.md`). No framework file was modified; VERITAS remains the unmodified object under study.